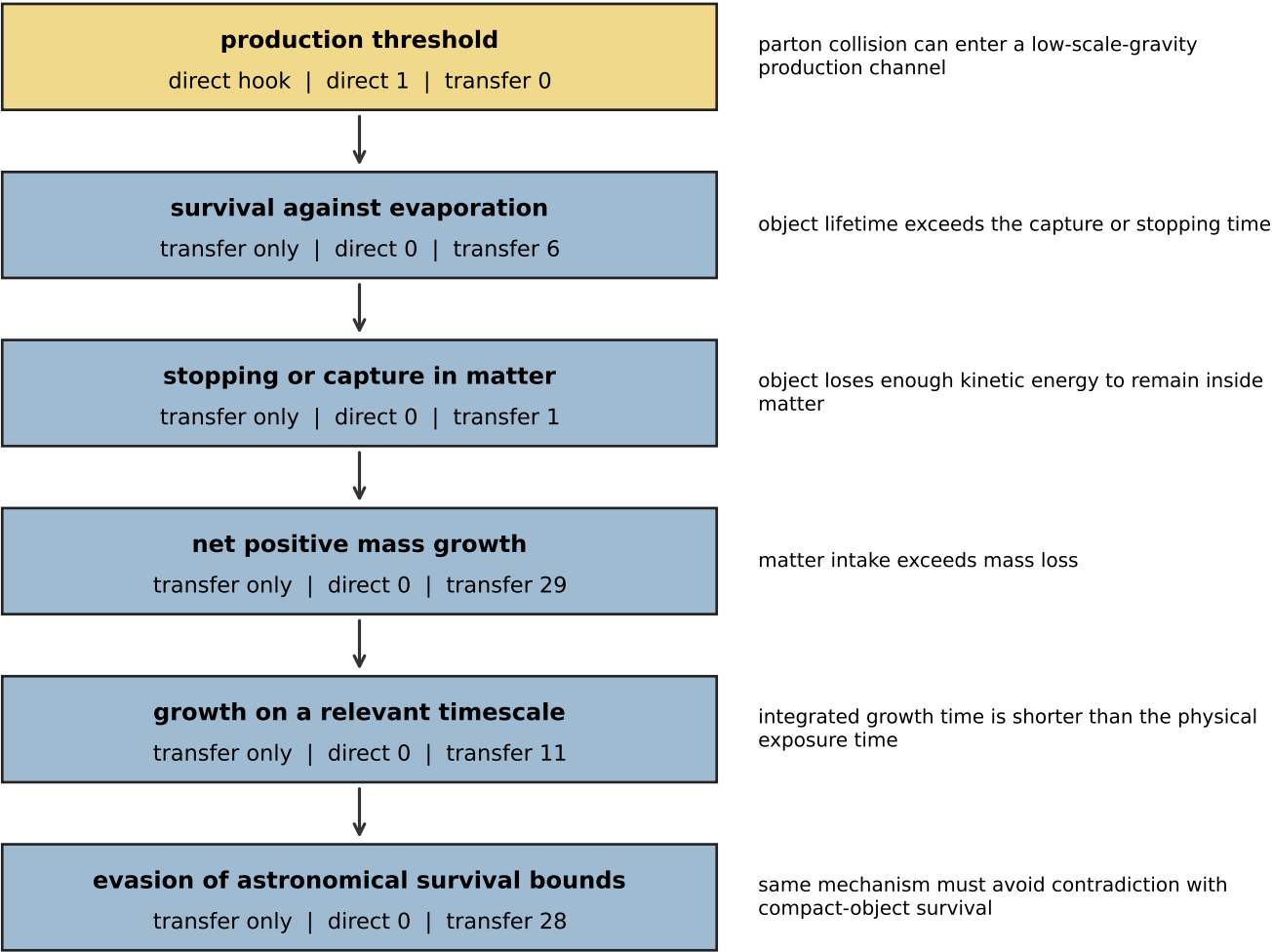


Required physical branch: every slot must be filled by collider-relevant equations



Result: the corpus contains a production hook and astrophysical transfer equations, but no direct collider chain for lifetime, capture and growth.